MACHINA-X: Explainable AI-Driven Digital Twin Framework for Predictive Maintenance

Comprehensive Empirical Research Report & Benchmark Evaluation

Project: MACHINA-X Digital Twin Platform
Research Domain: Cyber-Physical Systems, Electrical Machine Diagnostics, Explainable Artificial Intelligence (XAI)
Authors: Aditya Gupta, Aditya Maurya, Ananya Sharma, Aryan Mishra
Supervisor: Ms. Monika Sharma
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1. Executive Summary

This technical report delivers the complete empirical results, algorithmic benchmarks, and explainability evaluations for **MACHINA-X**, an explainable digital twin framework engineered for condition monitoring, anomaly detection, multi-fault diagnostics, and remaining useful life (RUL) estimation in industrial electro-mechanical systems.

The diagnostic pipeline is evaluated across two empirical tracks and one physical benchmark: 1. **Track A (UCI AI4I 2020 Benchmark)**: 10,000 synthetic milling-machine instances with extreme 28.5:1 class imbalance. 2. **Track B (Cyber-Physical Induction Motor Twin)**: 10,000 windows across 1,000 simulation runs modeling a 3 kW, 415 V, 50 Hz 4-pole motor across 5 health states. 3. **External Real-Data Physical Validation (CWRU Bearing Dataset)**: Real vibration accelerometer measurements from the Case Western Reserve University 12k Drive End testbed under cross-load evaluation.

2. System Architecture & Model Selection

- **Evaluation Partition for Selection**: VALIDATION (dataset_val.csv, zero test leakage)
- **Pareto Tolerance Band**: ≤ 0.02 Macro F1
- **Stage 1 Champion**: `IsolationForest`
- **Stage 2 Champion**: `XGBoost`

- **Decision Logic**: Top Validation Macro F1 is 0.9592. Candidates within 0.02 tolerance band: ['XGBoost', 'RandomForest']. 'XGBoost' is algorithmically selected based on validation performance (Val Macro F1 = 0.9558, Val Accuracy = 95.60%) while offering the lowest inference latency (1.17 ms mean, 1.04 ms p50) and native exact TreeSHAP explainability.

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3. Track A Benchmark: UCI AI4I 2020 Predictive Maintenance Dataset

- **Total Instances**: 10,000 (9,661 healthy, 339 failure, imbalance 28.5:1)
- **Champion Architecture**: XGBoost Classifier

Model Architecture	Test Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression (Baseline)	85.75%	17.80%	88.24%	0.2963	0.9393
Decision Tree (Baseline)	95.00%	38.89%	82.35%	0.5283	0.8568
Random Forest	98.40%	74.32%	80.88%	0.7746	0.9831
XGBoost Classifier	98.45%	74.03%	83.82%	0.7862	0.9808

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4. Track B Benchmark: Three-Phase Induction Motor Digital Twin

4.1 Dataset Semantics & Partitioning Protocol

- **Total Workload**: 10,000 1.0 s windows across 1,000 runs (10 windows/run)
- **Train Partition**: 6,000 windows (600 runs, 60.0%)
- **Validation Partition**: 2,000 windows (200 runs, 20.0%)
- **Held-Out Test Partition**: 2,000 windows (200 runs, 20.0%)

4.2 Stage 1 Anomaly Detection Performance

- **Isolation Forest (Champion)**: FPR = 4.25%, Recall = 70.38%, F1 = 0.8210, ROC-AUC = 0.8461
- **Autoencoder Baseline**: Recall = 97.44%, F1 = 0.8919, ROC-AUC = 0.8556

4.3 Stage 2 Multiclass Fault Classification (2,000 Unseen Test Windows)

- **XGBoost Classifier**: Test Accuracy = **96.40%**, Macro F1 = **0.9639**

Health / Fault Class	Precision (%)	Recall (%)	F1-Score	Test Windows Support
HEALTHY	96.59%	85.00%	0.9043	400
BROKEN_ROTOR_BAR	100.00%	100.00%	1.0000	400

Health / Fault Class	Precision (%)	Recall (%)	F1-Score	Test Windows Support
STATOR_INTER_TURN_SHORT	100.00%	100.00%	1.0000	400
BEARING_DEFECT	86.64%	97.25%	0.9164	400
AIR_GAP_ECCENTRICITY	100.00%	99.75%	0.9987	400

4.4 Multi-Seed Robustness & Leave-One-Load-Out Generalization

- **5-Seed Cross-Run Stability**: Accuracy = 95.30% $\pm$ 0.71%, Macro F1 = 0.9528 $\pm$ 0.0071
- **Leave-One-Load-Out (LOLO)**: Mean Accuracy across loads = 70.42%, Mean Macro F1 = 0.6572

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5. Remaining Useful Life (RUL) Prognostics

- **Evaluated Sequences**: 1,152 sequences across 12 held-out degradation runs (history length $k=5$)
- **Champion LSTM Regressor**: MAE = **0.7466 hours**, RMSE = **1.1849 hours**
- **Gradient Boosting Baseline**: MAE = 0.8172 hours, RMSE = 1.2348 hours
- **Ridge Baseline**: MAE = 1.2986 hours, RMSE = 1.7571 hours

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6. External Real-Data Physical Validation: Case Western Reserve University (CWRU)

- **Testbed**: Reliance Electric 2 HP Induction Motor with Deep Groove Ball Bearings (6205-2RS JEM SKF)
- **Sampling & Filtering**: Resampled from 12,000 Hz to 10,000 Hz polyphase FIR
- **Cross-Load Evaluation (Train Loads 0-1, Test Load 3)**: Accuracy = **70.00%**, Macro F1 = **0.3626** (70 test windows)
- Normal Class Recall: 100.0%, Precision: 93.02%
- Inner Race Recall: 90.00%, Precision: 33.33%
- **Few-Shot Fine-Tuning Transfer Progression**:
- $k = 0$: Mean Accuracy = 57.14% ($\pm 0.00\%$), Macro F1 = 0.3636
- $k = 5$: Mean Accuracy = 57.14% ($\pm 0.00\%$), Macro F1 = 0.1818
- $k = 10$: Mean Accuracy = 81.57% ($\pm 4.63\%$), Macro F1 = 0.6321
- $k = 20$: Mean Accuracy = 75.43% ($\pm 7.25\%$), Macro F1 = 0.4933
- $k = 50$: Mean Accuracy = 79.29% ($\pm 6.07\%$), Macro F1 = 0.5792

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7. Hardware Inference Latency Benchmark

Directly timed over **500 consecutive 1-second telemetry windows** on host hardware (Windows 10 (10.0.26200), AMD Ryzen 5 4600H with Radeon Graphics, Python 3.10.0): - **Physics Feature Extraction (26 features)**: Mean = 2.75 ms | p50 = 2.66 ms | p95 = 3.47 ms - **Stage 1 Isolation Forest**: Mean = 10.90 ms | p50 = 10.71 ms | p95 = 12.02 ms - **Stage 2 XGBoost Classifier**: Mean = 1.17 ms | p50 = 1.04 ms | p95 = 1.70 ms - **Stage 4 TreeSHAP Attribution**: Mean = 4.80 ms | p50 = 4.67 ms | p95 = 6.06 ms - **Complete End-to-End Pipeline**: Mean = **32.80 ms** | p50 = **31.87 ms** | p95 = **38.00 ms**

8. Scientific Limitations & Research Boundaries

1. **Simulated RUL Trajectory Boundaries**: Stage 3 LSTM RUL predictions are evaluated across 12 held-out simulated degradation runs (1,152 test sequences, MAE = 0.7466 hrs, RMSE = 1.1849 hrs). While mathematically rigorous on the simulated multi-law degradation model, these results must not be claimed as physical motor remaining-life validation without multi-month physical endurance bench testing.
2. **CWRU Physical Generalization Bounds**: The 70.00% zero-shot cross-load accuracy on CWRU bearing vibration (tested on Load 3 after training on Loads 0-1) exhibits class imbalance: Normal recall is 100.0%, but fault mode shifts require few-shot real-window fine-tuning (reaching 81.57% at k=10) to accommodate load-dependent slip and contact angle alterations.
3. **Severe Class Imbalance in AI4I Benchmark**: The 98.45% test accuracy on UCI AI4I 2020 operates against a 96.61% healthy baseline. Model evaluation must emphasize the minority failure-class binary F1 (0.7862) and macro F1 (0.8891) rather than raw accuracy alone.
4. **Simulator vs Physical IoT Telemetry**: The live streaming motor telemetry in the demonstrator originates from an in-memory physical differential simulator; it demonstrates edge architecture and explainability readiness, not a deployed industrial field IoT sensor network.
5. **Inverter Carrier Harmonics**: The current physical model assumes pure sinusoidal line voltage; variable-frequency drive (VFD) carrier harmonics (2-16 kHz) remain an area for future hardware-in-the-loop validation.
6. **Compound Fault Coupling**: When electrical stator shorts and mechanical bearing defects occur concurrently, feature attributions reflect both fault vectors; out-of-distribution rejection routes cases with posterior entropy > 0.40 to human engineer review.